INVERTIS UNIVERSITY, BAREILLY

FACULTY OF ENGINEERING & TECHNOLOGY

OFFICIAL ACADEMIC LEARNING RESOURCE â¢ DEPARTMENT OF CSE

Course: [BCS201] Programming in C (Programming for Problem Solving)

Programme: Bachelor of Technology in Computer Science & Engineering **Resource Type: LECTURE NOTES**

Module / Unit: Unit 1 **Faculty Author: Prof. Neha Gupta (Assistant P**

Unit 1 & 2 Comprehensive Notes: C Syntax, Control Structures & Loops

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1. C Language Fundamentals & Translation Pipeline

- Compilation Phases: Preprocessing (#include, #define), Compilation, Assembly, Linking.
- Primitive Data Types: int, char, float, double with size ranges and signedness modifiers.
- Storage Classes: auto, register, static, extern with scope, lifetime, and default linkage.

2. Control Flow, Branching & Loop Optimization

- Conditional Branching: if-else ladders, nested decisions, switch-case fallthrough rules.
- Iterative Structures: entry-controlled (for, while) vs exit-controlled (do-while) loops.
- Flow Control: break, continue, return, and disciplined avoidance of unconditional goto.